3.3 Calibration Metrics

Comment: The cited paper proposes an improved version of ECE. Should cite Guo, Pleiss, Sun et al. 2017, Niculescu-Mizil and Caruana 2005, etc. for ECE

We evaluated model calibration using Expected Calibration Error (ECE) (Błasiok & Nakkiran, 2023), which measures the discrepancy between confidence estimates and actual accuracy. We also utilized reliability diagrams to visualize calibration performance.

Comment: Although the AI Scientist generated reliability diagrams during the experiments, they were never included in the paper.

3.4 Training Procedure

Models were trained using standard cross-entropy loss and stochastic gradient descent with momentum. We used an initial learning rate of 0.1, decayed by a factor of 0.1 at epochs 50 and 75, for a total of 100 epochs. The batch size was set to 128. We followed consistent training procedures across all experiments to ensure comparability. Additionally, we applied temperature scaling (Kull et al., 2019) as a post-hoc calibration method to assess its effectiveness under label noise.

Comment: Experiments use Cosine Annealing Schedule and not stepwise decay.

4 Experiments and Results

4.1 Impact of Label Noise on Calibration

We first analyzed how different noise types and levels affect model calibration on CIFAR-10.

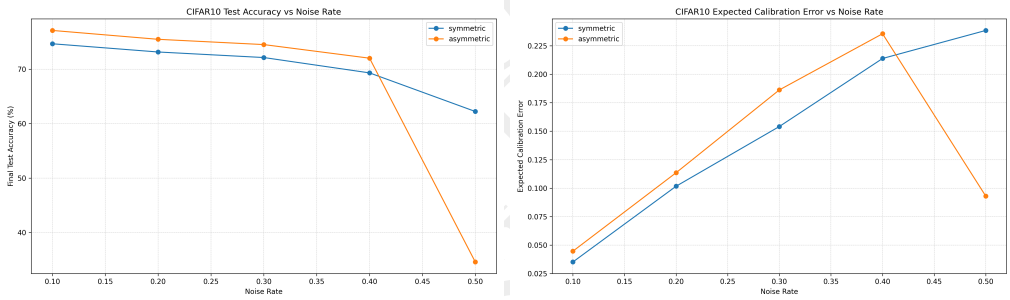

Figure 1: CIFAR-10 results: (Left) Test Accuracy vs. Noise Rate; (Right) ECE vs. Noise Rate for symmetric and asymmetric label noise.

Comment: The description of Figure 1 is not accurate. For example, the cited number (85%) is wrong, it should be 75%, and also should mention it's referring to 'symmetric'.

As shown in Figure 1, increasing label noise leads to a decline in test accuracy for both symmetric and asymmetric noise. Specifically, test accuracy drops from approximately 85% with no noise to around 60% at 50% noise rate. However, asymmetric noise has a more severe impact on calibration, with ECE increasing more rapidly compared to symmetric noise, reaching up to 0.35 at higher noise levels.

Comment: The statement that asymmetric noise has a more severe impact on calibration is inaccurate because the figure shows a more nuanced pattern–it first increases and then decreases after a noise rate of 0.4. Also the cited number (0.35) is incorrect; it should be 0.23 or 0.24.

4.2 Calibration Across Datasets

We extended the analysis to MNIST and Fashion-MNIST to assess whether the observed effects generalize across datasets.

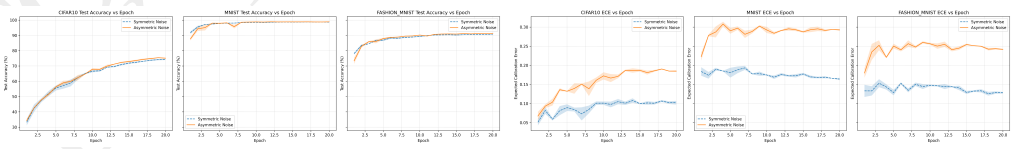

Figure 2: Test Accuracy (left) and ECE (right) over training epochs for CIFAR-10, MNIST, and Fashion-MNIST under symmetric and asymmetric label noise.

Comment: True for asymmetric noise, but would be better if symmetric noise results were discussed too.

Figure 2 shows that the negative impact of label noise on accuracy is consistent across datasets. Models trained on MNIST exhibit higher resilience in terms of accuracy, maintaining above 90% accuracy even at higher noise levels, but still suffer from increased ECE under asymmetric noise.